

Siman Shrestha

Software Engineer (Backend)

siman4457@gmail.com | (469)-323-7640 | Austin, TX | [LinkedIn](#) | [Portfolio](#)

Backend focused Software Engineer with 5+ years of experience building and scaling distributed systems in fintech and healthcare. Strong background in payments, data pipelines, and observability. Relocating to NYC.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript

Frameworks: Spring Boot, React, Node.js, Express, NestJS, REST, GraphQL

Databases & Streaming: MySQL, PostgreSQL, Kafka, Redis, MongoDB, Prometheus, Hazelcast

Cloud & Tools: AWS, Docker, Kubernetes, Jenkins, Grafana, Github Actions, Splunk, DataDog

AI & Tooling: LangChain, FastMCP, Streamlit, GitHub Copilot, Cline

EXPERIENCE

Visa | Austin, TX

Senior Backend Software Engineer

Sep 2024 – Present

- Built and maintained backend microservices for Visa Accept, designed to scale to **1M+** users and support tap and QR payments as the product expands to 50 countries using **Java**, **MySQL**, and **Hazelcast**.
- Led the design and development of a global transaction fees service, supporting Visa Accept's revenue model with FY26 projections of **\$2.5M**.
- Delivered APIs for merchant onboarding and transaction processing in close collaboration with iOS and product.
- Owned production service health and observability by instrumenting **Prometheus** metrics and **Grafana** dashboards, enabling automated failover, and global traffic routing.
- Developed a **Python**-based AI agent using GPT-4o and LangChain to surface merchant business insights from payment data, which received SVP endorsement and internal adoption by the product team.
- Built an **MCP** server with **FastMCP** to aggregate logs and Jira issues, reducing resolution time for incidents.

Hinge Health | Austin, TX

Backend Software Engineer II

Aug 2023 – Jul 2024

- Built and scaled microservices at a healthtech startup that manages sensitive user health data for 30M patients, enabling musculoskeletal care delivery at scale using **TypeScript**, **NestJS**, **Postgres**, and **AWS**.
- Accelerated targeted ad campaign delivery, drove a **6% increase in user registrations**, and **saved \$10k** annually on compute by evolving a hackathon prototype into a production **Kafka** data ingestion pipeline.
- Implemented **envelope encryption** for data at rest in **Kafka**, ensuring HIPAA and SOC 2 compliance.
- Built an asynchronous job processing system using BullMQ, Terraform, and **AWS** ElastiCache that **reduced peak load by 30%** and ensured accurate billing through record deduplication.

Visa | Austin, TX

Senior Fullstack Software Engineer

Oct 2020 – Aug 2023

- Built a real-time metrics dashboard used by **500+** reliability engineers, reducing query latency from 20s to 1s and accelerating production incident detection through **Redis** and **GraphQL** subscriptions over **WebSockets**.
- Owned and scaled an in-house observability platform used by **1000+** engineers to enable application monitoring, alerting, and logging with **React**, **Node.js**, and **MongoDB**.
- Led the multi-region rollout of **Grafana** Enterprise to support observability for mission-critical payment APIs, used by **1,000+** engineers across global teams.
- Implemented a full-stack job queuing platform to automate **300+** weekly tasks with **Angular**, **Java**, and **MySQL**.

EDUCATION

Texas A&M University, College Station, TX

May 2020

Bachelor of Science in Computer Science, Minor in Visualization